"breathing" mode. We would expect these fluctuations to be especially long-lived in the case of large values of μ. Decreasing the values of μ will increase the frequency of the oscillations; if the frequency of the periodic motion is too high then the overall stability of the simulation will become threatened (a smaller timestep h may be required).

Choosing $\gamma = 0.1$ or $\gamma = 1$ removes this artificial oscillatory behavior, with the larger choice of γ giving faster dissipation of the periodic oscillation. Figure 8.7 demonstrates that when setting $\gamma = 1$ the temperature and volume rapidly stabilize and give small, random fluctuations compared to the non-stochastic case. For some alternatives and more in-depth discussion of constant temperature and pressure strategies, including numerical experiments, refer to [128, 194, 199, 238, 252].

8.8.3 *Anisotropic Simulation Cell: The Method of Parrinello and Rahman*

Not all molecules are naturally simulated in a cubic unit cell and indeed the system may undergo dramatic changes of shape during simulation. Parrinello and Rahman [296] explained how to extend the method of Andersen to allow for a system confined to a cell with dimensions that are allowed to vary during simulation. Three vectors $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{c}$ define the sides of the simulation cell. Let $\boldsymbol{h}$ be the 3×3 matrix with these vectors as columns, then its volume is $\mathrm{V} = \det(\boldsymbol{h}) = \boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c})$, where $\boldsymbol{a}$, $\boldsymbol{b}$, $\boldsymbol{c}$ and $\boldsymbol{h}$ are viewed as time-dependent variables.

Any position vector may be decomposed into coordinates defined by the matrix h, thus

$$\boldsymbol{q}_i = \boldsymbol{h}\bar{\boldsymbol{q}}_i = \xi_i \boldsymbol{a} + \eta_i \boldsymbol{b} + \zeta_i \boldsymbol{c}, \qquad 0 \leq \xi_i, \eta_i, \zeta_i \leq 1.$$

Introducing the metric tensor $\boldsymbol{G} = \boldsymbol{h}^T \boldsymbol{h}$, the Lagrangian describing the system then becomes

$$L = \frac{1}{2} \sum_{i=1}^{\mathrm{N}} m_i \dot{\bar{\boldsymbol{q}}}_i^T \boldsymbol{G} \dot{\bar{\boldsymbol{q}}}_i - \bar{U}(\bar{\boldsymbol{q}}) + \frac{1}{2} \mu \mathrm{Tr}(\dot{\boldsymbol{h}}^T \dot{\boldsymbol{h}}) - \mathrm{PV},$$

where μ is an artificial "mass" parameter which can be adjusted to alter the tightness of the coupling of the barostat to the physical variables.

This system can then be coupled to a thermostat as in the Andersen or Hoover formulation [266].

The method works well as long as there are internal molecular structures that prevent a degeneracy (one cell dimension collapsing) as might happen, for example, if the scheme were applied to a simple liquid such as Lennard-Jonesium.